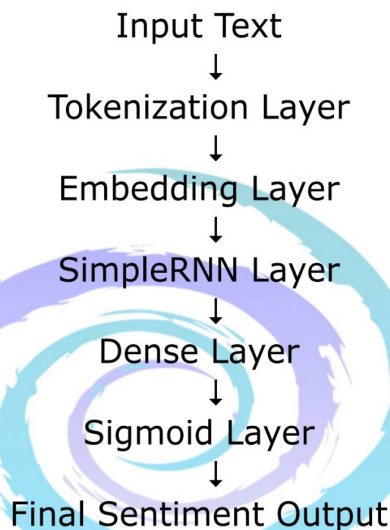


RNN Mathematical Calculations



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Layer 1: Input Layer

Input

food was not good

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This is raw text input.

The model cannot directly process words, so each word must be converted into a number.

Layer 2: Tokenization Layer

Token Table

food = 1
was = 2
good = 3
bad = 4
not = 5

Input Sentence

food was not good

Tokenized Output

[1, 2, 5, 3]

Meaning

food → 1
was → 2

not → 3
good → 3

So the sentence becomes:

[1, 2, 5, 3]

Layer 3: Embedding Layer

Purpose of Embedding Layer

The token numbers are only IDs.
They do not contain meaning.

Embedding layer converts each token ID into a vector.

Embedding Matrix

Token 0	Padding	[0.0, 0.0, 0.0]
Token 1	food	[0.2, 0.4, 0.1]
Token 2	was	[0.3, 0.1, 0.5]
Token 3	good	[0.8, 0.7, 0.9]
Token 4	bad	[0.1, 0.2, 0.9]
Token 5	not	[0.9, 0.3, 0.2]

Input to Embedding Layer

[1, 2, 5, 3]

Embedding Lookup

1 → [0.2, 0.4, 0.1]
2 → [0.3, 0.1, 0.5]
5 → [0.9, 0.3, 0.2]
3 → [0.8, 0.7, 0.9]

Output of Embedding Layer

X = [
[0.2, 0.4, 0.1],
[0.3, 0.1, 0.5],
[0.9, 0.3, 0.2],
[0.8, 0.7, 0.9]
]

Shape

Number of words = 4
Embedding dimension = 3

Shape = (4, 3)

Layer 4: SimpleRNN Layer

Purpose of RNN Layer

The RNN reads words one by one.

At every time step, it combines:

Current word vector
Previous hidden state
Bias

RNN Formula

$$h_t = \tanh(W_x \times x_t + W_h \times h_{t-1} + b)$$

Where:

h_t current hidden state
 x_t current word vector
 h_{t-1} previous hidden state
 W_x input-to-hidden weight matrix

Wh hidden-to-hidden weight matrix
b bias vector
tanh activation function

RNN Weights and Bias

Input-to-Hidden Weight Matrix: Wx

Wx = $\begin{bmatrix} [0.5, 0.1, 0.3], \\ [0.2, 0.4, 0.1], \\ [0.3, 0.2, 0.5] \end{bmatrix}$

Hidden-to-Hidden Weight Matrix: Wh

Wh = $\begin{bmatrix} [0.4, 0.2, 0.1], \\ [0.1, 0.5, 0.2], \\ [0.3, 0.1, 0.4] \end{bmatrix}$

Bias Vector: b

b = $[0.1, 0.1, 0.1]$

Initial Hidden State

h0 = $[0.0, 0.0, 0.0]$

Time Step 1: Word = food

Current Word

food

Current Input Vector

x1 = $[0.2, 0.4, 0.1]$

Previous Hidden State

h0 = $[0.0, 0.0, 0.0]$

$$h1 = \tanh(Wx \times x1 + Wh \times h0 + b)$$

Part 1: Calculate $Wx \times x1$

$$Wx = \begin{bmatrix} 0.5, 0.1, 0.3, \\ 0.2, 0.4, 0.1, \\ 0.3, 0.2, 0.5 \end{bmatrix}$$

$$x1 = [0.2, 0.4, 0.1]$$

Row 1

$$(0.5 \times 0.2) + (0.1 \times 0.4) + (0.3 \times 0.1)$$

$$= 0.10 + 0.04 + 0.03$$

$$= 0.17$$

Row 2

$$(0.2 \times 0.2) + (0.4 \times 0.4) + (0.1 \times 0.1)$$

$$= 0.04 + 0.16 + 0.01$$

$$= 0.21$$

Row 3

$$(0.3 \times 0.2) + (0.2 \times 0.4) + (0.5 \times 0.1)$$

$$= 0.06 + 0.08 + 0.05$$

$$= 0.19$$

Result

$$Wx \times x1 = [0.17, 0.21, 0.19]$$

Part 2: Calculate $Wh \times h0$

$$h0 = [0.0, 0.0, 0.0]$$

Since all values are zero:

$$Wh \times h0 = [0.0, 0.0, 0.0]$$

Part 3: Add Bias

$$Wx \times x1 = [0.17, 0.21, 0.19]$$

$$Wh \times h0 = [0.00, 0.00, 0.00]$$

$$b = [0.10, 0.10, 0.10]$$

$$\text{Total} = [0.17 + 0.00 + 0.10, \\ 0.21 + 0.00 + 0.10, \\ 0.19 + 0.00 + 0.10]$$

$$\text{Total} = [0.27, 0.31, 0.29]$$

Part 4: Apply tanh

$$h1 = \tanh([0.27, 0.31, 0.29])$$

$$\tanh(0.27) = 0.2636$$

$$\tanh(0.31) = 0.3004$$

$$\tanh(0.29) = 0.2821$$

Output of Time Step 1

$$h1 = [0.2636, 0.3004, 0.2821]$$

Meaning

Memory after reading "food"

Time Step 2: Word = was

Current Word

Current Input Vector

$$x2 = [0.3, 0.1, 0.5]$$

Previous Hidden State

$$h1 = [0.2636, 0.3004, 0.2821]$$

Formula

$$h2 = \tanh(Wx \times x2 + Wh \times h1 + b)$$

Part 1: Calculate $Wx \times x2$

$$x2 = [0.3, 0.1, 0.5]$$

Row 1

$$\begin{aligned} & (0.5 \times 0.3) + (0.1 \times 0.1) + (0.3 \times 0.5) \\ &= 0.15 + 0.01 + 0.15 \\ &= 0.31 \end{aligned}$$

Row 2

$$\begin{aligned} & (0.2 \times 0.3) + (0.4 \times 0.1) + (0.1 \times 0.5) \\ &= 0.06 + 0.04 + 0.05 \\ &= 0.15 \end{aligned}$$

Row 3

$$\begin{aligned} & (0.3 \times 0.3) + (0.2 \times 0.1) + (0.5 \times 0.5) \\ &= 0.09 + 0.02 + 0.25 \\ &= 0.36 \end{aligned}$$

Result

$$Wx \times x2 = [0.31, 0.15, 0.36]$$

Part 2: Calculate $Wh \times h1$

$$h1 = [0.2636, 0.3004, 0.2821]$$

Row 1

$$\begin{aligned} & (0.4 \times 0.2636) + (0.2 \times 0.3004) + (0.1 \times 0.2821) \\ &= 0.1054 + 0.0601 + 0.0282 \\ &= 0.1937 \end{aligned}$$

Row 2

$$\begin{aligned} & (0.1 \times 0.2636) + (0.5 \times 0.3004) + (0.2 \times 0.2821) \\ &= 0.0264 + 0.1502 + 0.0564 \\ &= 0.2330 \end{aligned}$$

Row 3

$$\begin{aligned} & (0.3 \times 0.2636) + (0.1 \times 0.3004) + (0.4 \times 0.2821) \\ &= 0.0791 + 0.0300 + 0.1128 \\ &= 0.2219 \end{aligned}$$

Result

$$Wh \times h1 = [0.1937, 0.2330, 0.2219]$$

Part 3: Add Bias

$$\begin{aligned} Wx \times x2 &= [0.3100, 0.1500, 0.3600] \\ Wh \times h1 &= [0.1937, 0.2330, 0.2219] \\ b &= [0.1000, 0.1000, 0.1000] \end{aligned}$$

$$\begin{aligned} \text{Total} &= [0.3100 + 0.1937 + 0.1000, \\ & \quad 0.1500 + 0.2330 + 0.1000, \\ & \quad 0.3600 + 0.2219 + 0.1000] \end{aligned}$$

$$\text{Total} = [0.6037, 0.4830, 0.6819]$$

Part 4: Apply tanh

$$h2 = \tanh([0.6037, 0.4830, 0.6819])$$

$$\begin{aligned} \tanh(0.6037) &= 0.5397 \\ \tanh(0.4830) &= 0.4486 \\ \tanh(0.6819) &= 0.5928 \end{aligned}$$

Output of Time Step 2

$h_2 = [0.5397, 0.4486, 0.5928]$

Meaning

Memory after reading "food was"



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Time Step 3: Word = not

Current Word

not

Current Input Vector

$x_3 = [0.9, 0.3, 0.2]$

Previous Hidden State

$h_2 = [0.5397, 0.4486, 0.5928]$

Formula

$h_3 = \tanh(W_x \times x_3 + W_h \times h_2 + b)$

Part 1: Calculate $W_x \times x_3$

$x_3 = [0.9, 0.3, 0.2]$

Row 1

$(0.5 \times 0.9) + (0.1 \times 0.3) + (0.3 \times 0.2)$
 $= 0.45 + 0.03 + 0.06$
 $= 0.54$

Row 2

$(0.2 \times 0.9) + (0.4 \times 0.3) + (0.1 \times 0.2)$
 $= 0.18 + 0.12 + 0.02$
 $= 0.32$

Row 3

$(0.3 \times 0.9) + (0.2 \times 0.3) + (0.5 \times 0.2)$

$$= 0.27 + 0.00 + 0.16$$

$$= 0.43$$

Result

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$$Wx \times x3 = [0.54, 0.32, 0.43]$$



Part 2: Calculate $Wh \times h2$

$$h2 = [0.5397, 0.4486, 0.5928]$$

Row 1

$$(0.4 \times 0.5397) + (0.2 \times 0.4486) + (0.1 \times 0.5928)$$

$$= 0.2159 + 0.0897 + 0.0593$$

$$= 0.3649$$

Row 2

$$(0.1 \times 0.5397) + (0.5 \times 0.4486) + (0.2 \times 0.5928)$$

$$= 0.0540 + 0.2243 + 0.1186$$

$$= 0.3969$$

Row 3

$$(0.3 \times 0.5397) + (0.1 \times 0.4486) + (0.4 \times 0.5928)$$

$$= 0.1619 + 0.0449 + 0.2371$$

$$= 0.4439$$

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Result

$$Wh \times h2 = [0.3649, 0.3969, 0.4439]$$

Part 3: Add Bias

$$Wx \times x3 = [0.5400, 0.3200, 0.4300]$$

$$Wh \times h2 = [0.3649, 0.3969, 0.4439]$$

$$b = [0.1000, 0.1000, 0.1000]$$

$$\text{Total} = [0.5400 + 0.3649 + 0.1000,$$

$$0.3200 + 0.3969 + 0.1000,$$

$$0.4300 + 0.4439 + 0.1000]$$

$$\text{Total} = [1.0049, 0.8169, 0.9739]$$

Part 4: Apply tanh

$$h3 = \tanh([1.0049, 0.8169, 0.9739])$$

$$\tanh(1.0049) = 0.7636$$

$$\tanh(0.8169) = 0.6733$$

$$\tanh(0.9739) = 0.7505$$



Output of Time Step 3

$$h3 = [0.7636, 0.6733, 0.7505]$$

Meaning

Memory after reading "food was not"

Important point:

The word "not" changes the sentiment direction.

Time Step 4: Word = good

Current Word

good

Current Input Vector

$x_4 = [0.8, 0.7, 0.9]$

Previous Hidden State

$h_3 = [0.7636, 0.6733, 0.7505]$

Formula

$h_4 = \tanh(Wx \times x_4 + Wh \times h_3 + b)$

Part 1: Calculate $Wx \times x_4$

$x_4 = [0.8, 0.7, 0.9]$

Row 1

$(0.5 \times 0.8) + (0.1 \times 0.7) + (0.3 \times 0.9)$
 $= 0.40 + 0.07 + 0.27$
 $= 0.74$

Row 2

$(0.2 \times 0.8) + (0.4 \times 0.7) + (0.1 \times 0.9)$
 $= 0.16 + 0.28 + 0.09$
 $= 0.53$

Row 3

$(0.3 \times 0.8) + (0.2 \times 0.7) + (0.5 \times 0.9)$
 $= 0.24 + 0.14 + 0.45$
 $= 0.83$

Result

$Wx \times x_4 = [0.74, 0.53, 0.83]$

Part 2: Calculate $Wh \times h_3$

$h_3 = [0.7636, 0.6733, 0.7505]$

Row 1

$(0.4 \times 0.7636) + (0.2 \times 0.6733) + (0.1 \times 0.7505)$
 $= 0.3054 + 0.1347 + 0.0751$
 $= 0.5152$

Row 2

$$(0.1 \times 0.7636) + (0.1 \times 0.6733) + (0.4 \times 0.7505) \\ = 0.0764 + 0.0673 + 0.3002 \\ = 0.5631$$

Row 3

$$(0.3 \times 0.7636) + (0.1 \times 0.6733) + (0.4 \times 0.7505) \\ = 0.2291 + 0.0673 + 0.3002 \\ = 0.5966$$

Result

$$Wh \times h3 = [0.5152, 0.5631, 0.5966]$$

Part 3: Add Bias

$$Wx \times x4 = [0.7400, 0.5300, 0.8300] \\ Wh \times h3 = [0.5152, 0.5631, 0.5966] \\ b = [0.1000, 0.1000, 0.1000]$$

$$\text{Total} = [0.7400 + 0.5152 + 0.1000, \\ 0.5300 + 0.5631 + 0.1000, \\ 0.8300 + 0.5966 + 0.1000]$$

$$\text{Total} = [1.3552, 1.1931, 1.5266]$$

Part 4: Apply tanh

$$h4 = \tanh([1.3552, 1.1931, 1.5266])$$

$$\tanh(1.3552) = 0.8752 \\ \tanh(1.1931) = 0.8314 \\ \tanh(1.5266) = 0.9099$$

Output of Time Step 4

$$h4 = [0.8752, 0.8314, 0.9099]$$

Meaning

Final memory after reading complete sentence:
food was not good

Hidden State Summary

$h_0 = [0.0000, 0.0000, 0.0000]$

After food:

$h_1 = [0.2636, 0.3004, 0.2821]$

After was:

$h_2 = [0.5397, 0.4486, 0.5928]$

After not:

$h_3 = [0.7636, 0.6733, 0.7505]$

After good:

$h_4 = [0.8752, 0.8314, 0.9099]$

Traversal diagram

food



$h_1 = [0.2636, 0.3004, 0.2821]$

was



$h_2 = [0.5397, 0.4486, 0.5928]$

not



$h_3 = [0.7636, 0.6733, 0.7505]$

good



$h_4 = [0.8752, 0.8314, 0.9099]$

Layer 5: Dense Layer (Fully Connected)

Purpose of Dense Layer

The RNN produces final memory.

The Dense layer converts this final memory into a single score.

Input to Dense Layer

$h_4 = [0.8752, 0.8314, 0.9099]$

Dense Layer Weights

$W_y = [-2.0, -1.5, -1.2]$

Dense Layer Bias

$b_y = 1.0$

Dense Formula

$z = W_y \times h_4 + b_y$

Expanded:

$z = (w_1 \times h_{4_1}) + (w_2 \times h_{4_2}) + (w_3 \times h_{4_3}) + b_y$

Substitution:

$z = (-2.0 \times 0.8752) + (-1.5 \times 0.8314) + (-1.2 \times 0.9099) + 1.0$

Calculation:

$-2.0 \times 0.8752 = -1.7504$

$-1.5 \times 0.8314 = -1.2471$

$-1.2 \times 0.9099 = -1.0919$

Addition:

$z = -1.7504 + -1.2471 + -1.0919 + 1.0$

$z = -3.0894$

Output of Dense Layer

$$z = -3.0894$$

Layer 6: Sigmoid Layer

Purpose of Sigmoid Layer

The Dense layer gives a raw score.
Sigmoid converts the raw score into probability.

Sigmoid Formula

$$\text{sigmoid}(z) = 1 / (1 + e^{(-z)})$$

Input

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$$z = -3.0894$$

Calculation

$$\text{sigmoid}(-3.0894) = 1 / (1 + e^{(-(-3.0894))})$$

$$\text{sigmoid}(-3.0894) = 1 / (1 + e^{3.0894})$$

Approximate value:

$$e^{3.0894} \approx 21.965$$

So:

$$\text{sigmoid}(-3.0894) = 1 / (1 + 21.965)$$

$$\text{sigmoid}(-3.0894) = 1 / 22.965$$

$$\text{sigmoid}(-3.0894) \approx 0.0435$$

Output of Sigmoid Layer

Prediction value = 0.0435

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Layer 7: Final Decision Layer

Decision Rule

Prediction $\geq 0.5 \rightarrow$ Positive Sentiment
Prediction $< 0.5 \rightarrow$ Negative Sentiment

Prediction

$$0.0435 < 0.5$$

Final Output

Negative Sentiment

Sentiment Label

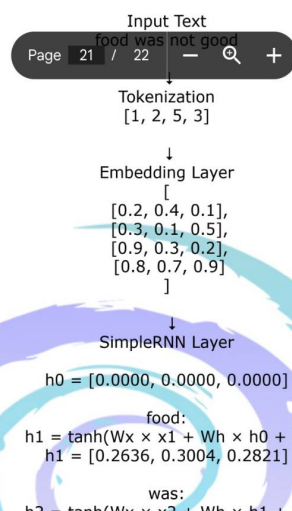
0

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Complete Layer-wise Input and Output Table

Layer	Input	Operation	Output
Input Layer	food was not good	Raw text input	food was not good
Tokenization	Words	Word to ID mapping	[1, 2, 5, 3]
Embedding	[1, 2, 5, 3]	Lookup vectors	(4, 3) matrix
RNN Time Step 1	food vector + h0	tanh calculation	h1
RNN Time Step 2	was vector + h1	tanh calculation	h2
RNN Time Step 3	not vector + h2	tanh calculation	h3
RNN Time Step 4	good vector + h3	tanh calculation	h4
Dense Layer	h4	Weighted sum	$z = -3.0894$
Sigmoid	z	Probability conversion	0.0435
Decision	0.0435	Threshold comparison	Negative

Complete Traversal Diagram



$$h3 = \tanh(Wx \times x3 + Wh \times h2 + b)$$
$$h3 = [0.7636, 0.6733, 0.7505]$$

good:

$$h4 = \tanh(Wx \times x4 + Wh \times h3 + b)$$

$$h4 = [0.8752, 0.8314, 0.9099]$$

↓

Dense Layer
 $z = W y \times h_4 + b_y$
 $z = -3.0894$

↓

Sigmoid Layer
prediction = sigmoid(z)
prediction = 0.0435

↓

Final Decision
 $0.0435 < 0.5$

↓

Negative Sentiment